

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: _____ WANG Kaiyuan _____

Student ID: _____ 22101552d _____

Part 2: Configure and OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1 ~~0~~ **0**

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 65 (serial link cost 64 + Ethernet cost 1) ~~129~~ **0**

Does R2 show up as an OSPF neighbor on R1? No ~~0~~ **0**

Does R2 show up as an OSPF neighbor on R3? No ~~2~~ **2**

What does this information tell you?

This information tells you that OSPF neighbor relationships between R2 and both R1 and R3 are not established, preventing the 192.168.2.0/24 network from being learned dynamically via OSPF, even though R3 has a directly connected path to it. ~~2~~

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1 **2**

R2(config-router)# no passive-interface s0/0/1 **2**

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1 **2**

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 (serial link cost 64 + Ethernet cost 1) **2**

Is R2 listed as an OSPF neighbor to R3? Yes **2**

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default OSPF reference bandwidth is 100 Mb/s, which is outdated for modern networks. Changing it to a higher value ensures that fast interfaces like Gigabit Ethernet receive accurate, differentiated costs rather than being grouped together with slower links.____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The costs were calculated using the OSPF formula $\text{Cost} = 100 \text{ Mbps} / \text{Interface Bandwidth}$, resulting in a cost of 64 for each serial link, which sum to 64 for the directly connected 192.168.23.0/30 network and 64 for the path to 192.168.3.0/24 via R3.____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost is 845 because the bandwidth 128 command changed the serial interface bandwidth to 128 Kbps, causing OSPF to recalculate the cost to approximately 781 for that link, which adds to the existing downstream costs.____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The ip ospf cost 1565 command increases the metric on R1's Serial0/0/1 interface to 1565, making the direct path to R3 much more expensive compared to the alternative route via R2. Since OSPF always selects the path with the lowest cumulative cost, it now prefers the less expensive path through R2 to reach the 192.168.3.0/24 network.____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

__ The router ID serves as a unique identifier for each OSPF router, playing a critical role in establishing and maintaining neighbor adjacencies. If Router IDs are not carefully managed and conflicts occur, it can result in incorrect routing information and instability within the OSPF domain.__

2. Why is the DR/BDR election process not a concern in this lab?

__ In this lab, the connections between routers are primarily point-to-point serial links. The Designated Router and Backup Designated Router election mechanism only applies to multi-access broadcast networks. Therefore, in this simplified topology, DR/BDR election does not take place and is irrelevant to the configuration.__

3. Why would you want to set an OSPF interface to passive?

__ Configuring an OSPF interface as passive prevents it from sending Hello packets and attempting to form neighbor adjacencies. This practice reduces unnecessary routing protocol traffic on local area networks (LANs) that do not require dynamic routing exchanges, while still allowing the network connected to that interface to be advertised to other OSPF routers.__

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

__ OSPF achieves its hierarchical design through the use of areas. By dividing a large autonomous system into smaller, interconnected areas—all of which must connect to the backbone area — OSPF limits the propagation of Link-State Advertisements and reduces the computational load on routers, enabling the protocol to scale effectively.__

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

router ospf 1

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

__ The router selects the route based on the Administrative Distance, not the metric, when routes to the same destination are learned from different routing protocols. EIGRP has a default AD of 90, OSPF has an AD of 110, and RIP has an AD of 120. Since EIGRP possesses the lowest AD value (90), it is considered the most trustworthy and will be installed in the routing table, regardless of the individual protocol metrics.__